Screening Cartels Under Incomplete Observability: Award-Layer Detection and the Architecture of Enforcement

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Abstract

Cartel-detection methods in the empirical industrial-organization literature read collusion off the distribution of submitted bids. Most enforcement environments globally preserve only the contract- award layer. We ask what cartel-screening properties survive that information coarsening, and we argue that the question reorganizes how enforcement should be architected. A separating-equilibrium framework with cover bidders identifies $\log(1 + \text{tenders\_count})$ together with the $\text{wins} = 0$ filter as a sufficient ranking statistic for cover-bidder type given an award-record envelope; the binary *frequent-loser* rule is its information-coarsening. In São Paulo's electronic procurement platform (2009–2019), the construct discriminates adjudicated cartel-cobidder firms inside the always-loser stratum at AUC 0.864 under temporal holdout (train 2009–2016, test 2017–2019) and at 0.748 on a strict pre-2020 benchmark with participation-stratified permutation null. Against bid-distribution detectors that require per-bidder bid amounts, the screening statistic reaches comparable accuracy on a substantially thinner data envelope (AUC 0.903 vs. 0.888 on the same target) and adds non-redundant signal in same-sample combination ($++0.035$ AUC, DeLong $p = 0.014$). Cobidders match the modeled cover-bidder type along four of five operational predictions (Cohen's d up to $+1.00$ within the always-loser stratum). The construct also leaves a $+3.6\%- + 7.7\%$ conditional log-price imprint as descriptive corroboration. The contribution is informational and architectural: an award-layer screening stage that prioritizes environments for the bid-layer forensic stage

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the prior literature requires, deployable wherever award records are preserved.

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1. Introduction

The empirical industrial-organization literature on cartel detection reads collusion off the distribution of submitted bids. Within-tender variance, kurtosis, skewness, and normalized spread are informative about coordinated bidding when the bid-level record is preserved (Porter and Zona, 1993; Bajari and Ye, 2003; Imhof, 2019; Wallimann et al., 2023). Real enforcement environments rarely match this benchmark. Most electronic procurement systems globally archive only the contract-award layer—winner identity, participant identity, item code, negotiated price—and export bid-by-bid records to operational logs that competition authorities and audit courts query slowly, partially, or not at all. The methods designed for the bid layer become inoperative on the layer that survives.

We treat the data-coarsening fact as a constraint on enforcement design and ask three connected questions. What collusion-relevant information remains in award-record data alone? Does that information separate cartel-adjacent populations from competitive ones with discrimination comparable to bid-distribution methods? And does the answer reorganize how an enforcement architecture under incomplete observability should be sequenced?

A separating-equilibrium framework with cover bidders and genuine entrants (Online Appendix A) makes the first two questions tractable. Two results anchor the empirical strategy. In equilibrium cover bidders satisfy $\Pr(\text{win} \mid C) = 0$: the wins $= 0$ filter is a separating choice, not a descriptive coincidence. And given award-record data, $\log(1 + \text{tenders\_count})$ together with the win-zero filter is a sufficient ranking statistic for cover-bidder type under a monotone-likelihood-ratio Poisson approximation. The framework treats endogenous sorting into cartel-affected items as constitutive of the screening object, not as a confound. The binary median $+ 1.5\times$IQR rule is its information-coarsening; we call firms above the threshold *frequent losers*.

The third question is informational and architectural. Under incomplete observability the appropriate enforcement design separates an *award-layer screening stage* that produces candidate environments for bid-layer scrutiny from a *bid-layer forensic stage* that adjudicates suspected coordination (Becker, 1968; Stigler, 1970; Harrington, 2008; Decarolis, 2014; ?). The two stages answer different questions, operate on different data envelopes, and trade off accuracy against deployability in different ways. The screening statistic this paper studies is the analytical object the award layer can deliver; whether the bid layer is preserved or not, the screening stage need not wait.

We test the framework on São Paulo's Bolsa Eletrônica de Compras (BEC, 2009–2019), an electronic platform that preserves the award envelope across 1,654,401 tender-items. Three discrimination results lead. First, against the cobidder population inside the always-loser stratum—the 193 firms that participated alongside adjudicated CADE direct defendants—the construct yields firm-level AUC 0.864 under temporal holdout (train 2009–2016, test 2017–2019) and 0.748 on a strict pre-2020 contemporaneous benchmark with participation-stratified permutation null (3.2× excess over the random-matching baseline, $p < 0.001$). Second, against bid-distribution detectors that require per-bidder bid amounts (Imhof 2019 pipeline, re-implemented; §10), the screening statistic reaches comparable accuracy on a substantially thinner data envelope (AUC 0.903 vs. 0.888 on the same target) and adds non-redundant signal in same-sample combination ($+ + 0.035$ AUC, DeLong $p = 0.014$). Third, against the 47 direct CADE defendants in the broader BEC firm universe, AUC is 0.49—the design's empirical signature, since the construct recovers loser-side participation rather than winner-side identity.

Within the always-loser stratum, the cobidder population matches the modeled cover-bidder type along four of five operational predictions the framework yields: cobidders deploy at higher intensity than FL non-cobidders (Cohen's $d = +1.00$ on unique winners faced), are deployed proximally to direct CADE defendants ($d = +0.46$), and concentrate their participation in narrower markets ($d = +0.39$ on portfolio item-group HHI). The validation positive class is therefore an empirically grounded analog of the modeled type, not a label of convenience.

The pricing evidence is descriptive corroboration. Frequent-loser presence associates with a $+3.6\% - +7.7\%$ higher conditional log negotiated unit price across four estimators (cross-fit, OLS, CEM, IPW). On the overlap subsample without ATT-reweighting the broad-sample positive imprint survives at $+0.044$ ($p = 0.035$); only ATT-weights that up-weight cells with relatively few untreated counterfactuals flip the sign to -9.72%. We report all three specifications and acknowledge that the framework's screening-value reading is one interpretation among several the data do not adjudicate (§7.2, Online Appendix A). Cinelli and Hazlett (2020)'s $RV_{q=1} = 17.5\%$ and Oster (2019)'s $\hat{\delta} = 261.6$ bound the broad-sample association against selection on observables.

The contribution sits at the intersection of four literatures, each constrained by the data envelope on which it operates. The bid-coordination literature (Bajari and Ye, 2003; Porter and Zona, 1993; Conley and Decarolis, 2016) requires a partition between colluders and a competitive fringe; the construct supplies a candidate *ex ante* firm-level flag computable from award records. Bid-distribution screens (Imhof, 2019; Wallimann et al., 2023; Huber and Imhof, 2019) require bid microdata; the construct operates on the information-coarsened layer that survives, and integrates with bid-distribution methods at the forensic stage where microdata exist (§10). Empirical cover-bidding work identifies cover bidders *ex post*, through leniency testimony or judicial discovery (Porter and Zona, 1999; Pesendorfer, 2000; Asker, 2010; **?**); we are not aware of a published prospective identification of cover-bidder candidates from public award-record data alone. The enforcement-design literature (Becker, 1968; Stigler, 1970; Harrington, 2008; Baker, 2003) prescribes how detection technology and information acquisition interact under costly observation; the screening-then-forensic sequencing this paper proposes is a particular structure of that information chain.

The cumulative claim is structural and informational. Under incomplete observability of the bid layer, an award-layer screening stage prioritizes environments at discrimination accuracy comparable to bid-distribution methods on a substantially thinner data envelope, with non-redundant signal in combination. The architectural implication is that the bid-layer forensic stage need not be the universal entry point for cartel enforcement;

jurisdictions that lack bid microdata can deploy the screening stage on data they already maintain, and reforms that mandate bid archival expand the forensic stage that the screening stage feeds rather than substitute for it.

Three claims the paper does not make. The pricing imprint is not a causal estimate of cover bidding's effect on prices, the screening framing does not require it to be, and the overlap-decomposition shows the sign reversal is a reweighting phenomenon that the data do not let us interpret causally. The buyer-size gradient is scope information about where the screening signal varies across detection regimes, not identification of an institutional channel. And the construct does not adjudicate cartel membership: cobidders are the validation object the data layer supports, and the AUC asymmetry against direct defendants is the design's empirical signature, not a failure of the screening logic.

The paper proceeds in five blocks. Section 2 situates the contribution against the bid-coordination, bid-distribution, cover-bidding, and enforcement-design literatures. Section 3 describes the BEC data layers and the CADE adjudication portfolio. Sections 4–5 state the participation primitive, the binary rule, and the empirical strategy, alongside the two design-based strategies (RDD, DiD) that return null. Section 6 carries the discrimination evidence and the within-stratum theory-validation bridge. Section 7 reports the price imprint as descriptive corroboration and decomposes the sign reversal under overlap restriction. Sections 8–9 develop the modal informativeness contrast, the detection-regime gradient, and the threshold/identification audits. Section 10 carries the architectural test against bid-distribution detectors trained on bid microdata. Sections 11–12 catalog scope limitations and conclude.

2. Related Literature

The paper sits at the conjunction of four literatures organized by the data layer at which each operates. Bid-coordination tests read collusion off bid-function exchangeability conditional on a researcher-supplied partition between colluders and a competitive fringe. Bid-distribution screens read collusion off within-tender moments of submitted bids. Cover-bidding studies identify the loser side of collusive arrangements retrospectively.

Work on enforcement design under incomplete information asks how detection regimes should be sequenced when monitoring is imperfect. The construct's positioning is determined by the informational layer at which each literature operates.

Bid-coordination tests presuppose a partition. Bajari and Ye (2003) establish the canonical framework for detecting collusion through exchangeability of bid functions and conditional independence of residuals, conditional on a researcher-supplied partition between suspected colluders and a competitive fringe. The papers that build on Bajari–Ye inherit the constraint: Porter and Zona (1993, 1999) detect bid rotation conditional on assumed cartel membership; Baldwin et al. (1997) use post-prosecution records to identify timber-auction colluders; Clark et al. (2021) and Schurter (2020) study cartel dynamics with ex-post-known membership. The partition is constructed *ex post* or by assumption in every case. The construct supplies a candidate *ex ante* firm-level flag computable from award records, which bid-coordination tests can take as input where microdata exist.

Conley–Decarolis: the closest precedent in data parsimony. Conley and Decarolis (2016) is the closest precedent for participation-only collusion detection in the empirical literature. Operating on Italian highway-construction auctions where investigators had identified a focal cartel but lacked a complete membership list, they infer bidder coalitions from co-bidding network structure—clustering coefficients, repeated-pair density, and degree distributions in firm-by-firm participation graphs—which identify the cartel's group boundaries from participation data alone. Three differences delimit the present contribution. First, the object: Conley and Decarolis recover *group structure* (which firms belong to the same coalition given a focal investigation), while the construct produces *firm-level flags* on a screening basis, where no investigation has been opened. Second, the timing: their exercise operates on a *post-investigation* pool of suspected firms; the construct operates on the *ex ante* universe of always-loser participants without any investigation as input. Third, the empirical content: their analysis recovers within-cartel structure on the assumption that a cartel exists; the construct asks whether the participation footprint suggests a cartel-affected environment in the first place. The two methods are sequential, not substitutive: a screening stage that flags candidate environments feeds investigations within which Conley–Decarolis-style coalition recovery becomes available.

Bid-distribution screens require the bid layer. Imhof (2019); Imhof et al. (2018); Huber and Imhof (2019) and Wallimann et al. (2023) train machine-learning classifiers on tender-level bid statistics—within-tender coefficient of variation, kurtosis, skewness, normalized spread—reaching AUC above 0.85 against ground-truth European cartel cases. Abrantes-Metz et al. (2006) pioneer the variance screen; Chassang and Ortner (2022) provide moment-based theoretical foundations; Harrington (2008) surveys the field. Every implementation operates at the tender level and requires per-bidder bid amounts. Real enforcement environments often preserve only the award envelope; the construct runs on that envelope and integrates with bid-distribution screens at the forensic stage where microdata are recoverable (§10).

Cover-bidding identification has been retrospective. Porter and Zona (1999); Pesendorfer (2000); Asker (2010) and Marshall and Marx (2012) document cover bidders in prosecuted cartels through case records (Ohio milk, Italian highway, NYC stamps, and the sequence reviewed by Marshall and Marx); Athey et al. (2011) compare cover-bidding incentives across auction formats theoretically. In every empirical case the cover bidders are identified *ex post*, through leniency testimony, post-conviction records, or judicial discovery, and after the cartel has been adjudicated. We are not aware of a published prospective identification of cover-bidder candidates from public award-record data. The construct shifts the post-adjudication observation into a pre-investigation screening stage.

Enforcement design under incomplete information. Becker (1968) and Stigler (1970) establish the optimal-deterrence baseline: when monitoring is imperfect, the sanction-detection trade-off has a structural solution that depends on the informational technology available to the enforcer. Baker (2003) treats the optimal sequencing of investigative stages under information asymmetry between agency and target: investigations are costly, and the screening rule that sorts candidates into investigation should economize on data the agency observes at low marginal cost. Two implications travel to our setting. First, when bid-level observability is absent, the appropriate screening object is whatever statistic the collusive arrangement generates that survives the data-coarsening step—a question of identifiability under monitoring constraints rather than of detection power against a fixed informational benchmark. Second, screening and forensic stages are sequenced, not parallel: the screening stage produces candidate environments for the forensic stage, which then deploys the more data-intensive bid-distribution moment battery (Chassang and Ortner, 2022; Harrington, 2008). The architectural separation between an award-layer screening stage and a bid-layer forensic stage that the construct instantiates is the empirical implication of this sequencing logic.

Procurement institutions and the observability environment. A growing empirical literature documents how procurement institutions shape what enforcement can observe. Coviello and Gagliarducci (2017) show that longer political tenure worsens procurement outcomes in Italy; Decarolis et al. (2020) show that discretionary procedures and limits to competition jointly predict corruption risk; Decarolis and Giuffrida (2017) documents the procurement-officer channel. We treat these institutional features as variation in the monitoring regime that the construct's signal is read against (§7.3).

Synthesis. The contribution is the missing screening-stage object under incomplete observability. Bid-coordination tests (Bajari and Ye, 2003) adjudicate inside an existing partition; the construct produces the firm-level candidate flag the tests take as input. Bid-distribution screens (Imhof, 2019; Wallimann et al., 2023) adjudicate inside the bid layer; the construct runs on the layer that survives when the bid layer does not, and the two methods are informational complements rather than substitutes (§10). Coalition-recovery methods (Conley and Decarolis, 2016) recover within-cartel structure once a cartel has been identified; the construct flags the candidate environments inside which such structure can be recovered. Under incomplete observability the participation footprint of cover bidders is itself an informative screening object, and the appropriate enforcement architecture sequences screening stages before forensic stages rather than treating them as competing methods.

3. Institutional Setting and the Observability Architecture

The empirical exercise rests on three institutional facts about the BEC enforcement environment, each load-bearing for the screening interpretation. Two procurement modalities expose different observability regimes, against which the construct's signal varies (§8.2). A data layer survives at the analytical warehouse while the bid-by-bid layer does not, defining the screening problem the paper takes as starting point. And a CADE adjudication portfolio supplies cartel-adjacency labels with a substantial gap between operational cartel activity and final ruling, which makes the validation in §6 partly prospective. The architecture is not Brazil-specific. Analogous regimes in EU and U.S. enforcement environments preserve a similar award/bid asymmetry, which we take up in the portability discussion of §11.3.

Two procurement modalities, two observability regimes. Brazilian public procurement during 2009–2019 is governed by Lei 8.666/93 and Lei 10.520/2002. Two modalities dominate BEC. *Convite* (Lei 8.666/93, Art. 22) is a sealed-bid invitation procedure with a statutory three-bidder quorum requirement and a low value cap; the only object preserved at the analytical-warehouse layer is the final award record. *Pregão* (Lei 10.520/2002) is a reverse electronic auction in which bidders submit progressively lower offers in real time through a public interface, with no minimum-bidder requirement; the real-time interaction footprint is observable to other bidders during the auction itself, and a richer set of bidder-by-bidder behaviors enters the operational record. The modalities therefore differ in two dimensions simultaneously: the bidding rule (sealed/quorum vs. real-time/no-quorum) and the observability regime (award-only vs. real-time-public). We treat the modal contrast as variation in the regime within which the screening signal is recovered, not as the mechanism the construct exploits. The 2021 reform (Lei 14.133/2021) postdates our sample and consolidates pregão-style auctions as the default, shifting the modal mix toward the regime in which the construct is most informative.

The data layer that survives at the analytical warehouse. The construct operates on the data layer that BEC's analytical warehouse exports: winner identity, participant identity, item code, negotiated price, procuring unit, and procurement-procedure metadata. Bid-by-bid records exist in the operational system but are not routinely exported to the layer that competition authorities and audit courts query.[2]

The asymmetry between the two layers is structural rather than parochial. The U.S. federal procurement system (FPDS-NG) reports contract awards and contracting-action history but does not preserve bid amounts at the contracting-officer-accessible layer; bid microdata, when retained at all, sit in agency-level solicitation files that the Federal Trade Commission and the Antitrust Division of the Department of Justice query through case-specific subpoenas rather than through routine analytical access. The European Union's Tenders Electronic Daily (TED) publishes the award-record layer mandatorily across member states but preserves bid microdata heterogeneously: a recent review of 12% procurement systems (Sánchez Graells, 2019) reports that fewer than half make per-bidder bid amounts retrievable at the analytical-warehouse layer. The U.K. Contracts Finder system preserves awards; bid-level archival depends on contracting-authority discretion. The screening problem the construct is built around is therefore a feature of the enforcement architecture as it actually exists in advanced and middle-income economies, not a Brazilian peculiarity. A screening object that operates on the layer universally preserved is, by construction, more deployable than one that requires the layer that is not.

Adjudication portfolio for validation labels. The Conselho Administrativo de Defesa Econômica (CADE) is the federal competition authority that adjudicates bid-rigging cases under Lei 12.529/2011, with leniency provisions (Art. 86) generating the disclosure trail the cartel-detection literature uses as ground truth. CADE's procurement-cartel portfolio relevant to our sample comprises 12 adjudicated cases covering 65 firm-defendants (47 active in BEC during the sample window), with final decisions spanning 2015–2025 and process-to-ruling intervals of 4–16 years. The interval is the relevant benchmark for the

[2]In TCE-SP audits we examined, requesting bid-level microdata from the BEC operational system is a multi-step administrative process that delays investigations by weeks; the award-record layer returns results in minutes. The latency cost has been flagged in OECD reviews of Brazilian procurement infrastructure (OECD, 2019).

screening-stage interpretation: an award-layer screen that flags candidate environments at t accelerates the path to forensic investigation relative to the post-adjudication identification on which the prior literature relies. The construction of the validation populations (direct defendants, cobidders inside the always-loser stratum) and the prospective-versus-contemporaneous benchmark distinction are reported in §6.

The administrative-audit institutions surrounding BEC (TCE-SP, CGE) supervise procurement legality and maintain debarred-firm registries; we do not require them as part of the empirical strategy and treat them as part of the broader monitoring environment within which the screening signal is read.

Together, the three institutional facts define the empirical setting within which the descriptive identifying stance of §5.1 is well-posed: the broad-sample association is computed on the layer that survives, the modal contrast reads informativeness across the two regimes that the layer preserves, and the validation labels are post-adjudication objects that the long process-to-ruling interval makes prospective on the screening side. The screening problem the construct addresses is therefore not a feature of the data we happen to have; it is a feature of the data layer that enforcement architectures of this type universally preserve.

4. Data and Frequent Losers Definition

4.1. BEC Platform

The data come from the Bolsa Eletrônica de Compras (BEC), São Paulo's centralized electronic procurement platform, operated by the State Treasury since 2002. The empirical setting is substantive in absolute terms rather than a peripheral national case: São Paulo state has 46.1 million residents (2025 IBGE estimate, 21.6% of Brazil's population) and a GDP of R$3.45 trillion in 2023, accounting for 31.5% of Brazilian GDP (IBGE

Sistema de Contas Regionais, released November 2025).[3] BEC serves 1,308 purchasing units (PBUs)—state agencies, public universities, hospitals, schools—and clears two procurement modalities during our sample: *convite* (sealed-bid invitation procedure with a statutory three-bidder requirement, Lei 8.666/93) and *pregão* (electronic reverse auction with no minimum-bidder requirement, Lei 10.520/2002). Our 2009–2019 window comprises 4,533,754 contract-record items across 22 calendar semesters, generated from 39,961,357 individual bids placed by 41,444 distinct firms (Table B.1). The contract-record layer preserves participants, winners, negotiated prices, reference prices, and procurement metadata; the bid-by-bid log is retained in the operational system but exported to the analytical warehouse only on administrative request, the data-layer asymmetry characterized in §3.

The modal mix shifts over the sample window. Pregão auctions expand from a minority share at the start of the sample toward near-universal coverage by 2019, consistent with the 2021 consolidation of pregão-style electronic auctions as the institutional default for Brazilian procurement (§3). Convite remains a meaningful share in the small-cap range and in early years, providing the within-platform variation that the modal contrast in §7.1 and §8.2 reads.

4.2. Frequent Losers Definition

Why a wins-zero filter. The framework's wins $= 0$ filter is a separating choice under standard expected-profit competition: a firm that bids without ever winning faces negative expected continuation profit once cumulative bidding cost exceeds the gain from any single plausible win. With per-bid cost c, mean order value $\bar{V}$, and gross margin m, the break-even on n losses is $n^* = \bar{V}m/c - 2$ (Online Appendix A). At BEC parameters ($\bar{V} \approx$ R\$86,000, $m = 0.10$, and the documented bid-preparation cost $c \in [\text{R\$200}, \text{R\$500}]$ for one to two staff-hours plus compliance), n^* falls in $[15, 41]$. The empirical FL

[3] At the average 2023 exchange rate of roughly R\$5/US\$, São Paulo's 2023 GDP is approximately US\$690 billion, which would place the state around the world's top 20 jurisdictions if treated as an independent economy—comparable to Switzerland, Argentina, or Sweden, and exceeding the GDP of Belgium, Norway, Austria, Ireland, or Israel (World Bank, 2023). State-level procurement spending in São Paulo is correspondingly substantial in absolute terms, which is the relevant scale for assessing the deployment value of an award-layer screening stage.

threshold of 14 tender-items lies in this interval; the 90^{th}-percentile FL participates in 50+ zero-win tenders and accumulates direct cost above R$10,000. Sustained zero-win participation past the break-even is consistent with cover bidding, capacity signaling, or eligibility maintenance; we organize the empirical work around the cover-bidding reading without claiming it as the unique interpretation (§11.2). The break-even calculation is a consistency check on the framework's wins = 0 filter, not an independent identification device.

Two-step classification. We define frequent losers as the intersection of two conditions:

1. **Always-losers**: firms with a zero win rate across all 2009–2019 BEC tenders. The pool comprises 16,843 firms (41% of all BEC participants); frequent losers are a small extreme subset of this stratum.
2. **IQR threshold**: among always-losers, those with participation counts above median + $1.5 \times$IQR of the always-loser participation distribution.[4] The threshold yields 14 or more tenders with zero wins, corresponding to a statistical cutoff of 13.5. The threshold was fixed *ex ante* on the participation distribution and the framework's break-even, before any examination of CADE labels. The definition yields 2,735 frequent-loser firms.

The treatment variable losers_{igt} equals one if tender-item i in item group g at time t has at least one frequent-loser firm among its participants, and zero otherwise.

The continuous primitive and its binary coarsening. The economic construct is broader than the binary rule. Proposition 2 identifies log(1+tenders_count) as a sufficient ranking statistic for cover-bidder type given award-record data; the binary 14-tender rule is the information-coarsening of this statistic under the framework's separating equilibrium. The horse race of §8.1 confirms the prediction empirically: the continuous statistic strictly dominates the binary in discrimination (DeLong et al. 1988 test, $p < 2e-05$). We use the

[4]We use median + 1.5× IQR rather than the Tukey rule (Q3 + 1.5× IQR) because the participation distribution is heavily right-skewed; the adopted rule yields a stricter threshold. The Tukey alternative flags 1,981 firms with AUC 0.834 ([0.804, 0.863]), against 0.924 ([0.921, 0.926]) under the adopted rule. Robustness to the multiplier choice between 1.0× and 3.0× is reported in §9.1.

continuous statistic as the discriminating instrument and the binary rule as the discrete cut-off the screening framing flags.

Out-of-sample generalization of the threshold. Pooling participation over the full 2009–2019 window maximizes within-stratum discrimination but does not isolate the out-of-sample stability of the screening statistic. Estimating the threshold on 2009–2016 alone gives a cutoff of 7 and a firm-level AUC of 0.767 ([0.734, 0.800]); evaluating the 2009–2016-frozen score against 2017–2019 items, the binary indicator drops to 0.565, while the continuous primitive retains 0.770 (Table 2). The full generalization audit and the implied conservative AUC of 0.864 are reported in §9.3.

Table 1: Loser-Side Concentration as Concept, FL as Operational Rule

Measure	Editorial role	Threshold	Full-sample AUC	Strict firm AUC	Strict item AUC
Continuous $\log(1 + \text{tc})$	Latent construct	—	0.939	0.750	0.770
Binary FL: median + 1.5 IQR	Operational rule	13.5	0.924	0.767	0.565
Alternative: Q3 + 1.5 IQR	Tukey alternative	18.5	0.834	—	—
Alternative: median + 1.0 IQR	Looser rule	10.0	0.902	—	—

Notes: The strict-train threshold estimated only on 2009–2016 is 7.0, far below the full-sample paper-rule threshold of 13.5. The framework identifies $\log(1 + \text{tenders\_count})$ as the participation primitive; the binary FL rule is its information-coarsening rather than a theoretically unique boundary.

Table 2: Strict Temporal Holdout with Threshold Frozen on 2009–2016

Scope	Score	Threshold	AUC	95% CI	Positives / N
Firm-level (AL)	Binary FL	7.0	0.767	$[0.734, 0.800]$	193/21,819
Firm-level (AL)	$\log(1 + \text{tc})$	—	0.750	$[0.706, 0.795]$	193/21,819
Item-level 17–19	Binary FL	7.0	0.565	$[0.564, 0.566]$	5,299/1,009,729
Item-level 17–19	$\log(1 + \text{tc})$	—	0.770	$[0.764, 0.776]$	5,299/1,009,729

Notes: Train-window threshold (2009–2016 AL): 7.0. Full-sample reference: 13.5. Binary FL threshold estimated only on 2009–2016; item-level rows apply the frozen threshold to 2017–2019 items.

4.3. Sample Construction and Descriptive Statistics

The participation distribution among always-losers is heavily right-skewed (Figures B.1–B.2). The regression sample retains convite and pregão tenders with a recorded winner and item types with at least one frequent-loser participation during the sample window.

The last restriction conditions on the treatment in the sense that item types that never attract frequent losers anywhere in 2009–2019 are excluded; this ensures that controls come from item markets where the screening statistic can plausibly fire. The unrestricted sample returns a coefficient within 0.5 percentage points of the restricted-sample baseline (§9), so the conditioning is empirically innocuous.

The regression sample contains 1,654,401 tender-items across 18,783 item types, defined at the finest product-code level available in BEC. Of the 18,783 item types, 15,101 (82%) have both frequent-loser-present and frequent-loser-absent tenders within sample, contributing to within-item identification under the fixed-effects specification of Equation (1). Frequent-loser firms participate in 4.8% of tender-items.

Table 3 reports unconditional descriptive statistics by frequent-loser presence. The raw price ratio is roughly elevenfold (R$140,673 vs R$12,465), reflecting the sorting of frequent losers toward larger contracts; frequent-loser-present tenders attract on average 9.06 bidders against 4.90 in frequent-loser-absent tenders. Under within-item, within-year, within-procuring-unit fixed effects, the 2.43-log-point raw gap compresses to roughly 0.05 log points (Table B.2). The selection that the fixed-effects structure absorbs is large; the residual within-item association is small in log points but systematically positive across estimators (§7.1).

Table 3: Descriptive Statistics: Tenders With vs. Without Frequent Losers

	With FL			Without FL		
	Mean	SD	N	Mean	SD	N
Negotiated price	140,673.37	3,148,893.84	79,452	12,465.30	484,167.26	1,574,949
Log negotiated price	4.92	3.87	79,452	2.49	2.59	1,574,949
Number of firms	9.06	6.35	79,456	4.90	3.17	1,574,991
Log number of firms	1.99	0.67	79,456	1.39	0.65	1,574,991
Number of bids	18.56	24.55	79,456	9.58	13.19	1,574,991
Non-FL firms	7.82	6.01	79,456	4.90	3.17	1,574,991
FL count per tender	1.25	0.75	79,456	0.00	0.00	1,574,991

Notes: Sample restricted to item types with $\geq$1 FL tender. FL = frequent losers defined by median + 1.5$\times$IQR threshold.

The CADE adjudication portfolio that supplies the validation labels is described in

§6.1, where the distinction between direct defendants and cobidders inside the always-loser stratum is the load-bearing structural feature.

5. Empirical Strategy

5.1. The Empirical Object: Discrimination of Cartel-Adjacent Populations Under Coarsened Observability

The primary empirical object is the discrimination accuracy of the participation-based statistic against cartel-adjacency labels under the data envelope the framework constrains us to: an award- record layer that preserves participant identity, winner identity, item code, and negotiated price, but not per-bidder bid amounts. The discrimination object is what the framework's sufficient ranking statistic predicts will dominate when only the award layer is observable, and it is the object the validation evidence in §6 carries.

The conditional log-price association reported in §7 is a secondary object: a descriptive corroboration that the screening signal also leaves a price imprint on the items it concentrates on, not a treatment-effect target. Under the separating equilibrium of Online Appendix A, frequent-loser participation is generated by the cartel's deployment problem; endogenous sorting into items where the cartel deploys is constitutive of the screening object. The screening question asks how informative the resulting participation footprint is about the collusive environment when only the award layer is observable; the relevant estimator integrates over the distribution of items into which frequent losers actually enter.

We therefore report two empirical objects, with the discrimination object as the primary defense and the conditional association as descriptive corroboration:

Primary object: discrimination AUC. the receiver-operating characteristic AUC of the binary frequent-loser indicator and the underlying continuous statistic $\log(1 + \text{tenders\_count})$ against three cartel-adjacency labels: (i) cobidders inside the always-loser stratum on a strict pre-2020 contemporaneous benchmark with participation-stratified permutation null (§6.3); (ii) cobidders under temporal holdout, train 2009–2016 and test 2017–2019 (§6.4); and (iii) the same target read against the Imhof (2019)–style bid-distribution pipeline trained on the bid layer (§10). The discrimination object is what the framework's sufficient ranking statistic predicts will dominate; the AUC numbers are the empirical content the screening interpretation rests on.

Secondary object: broad-sample conditional association (β). the within-cell conditional mean difference in log negotiated price between tender-items that include at least one frequent-loser participant and otherwise comparable items in the same product code, year, and procuring unit. We estimate β on 1,654,401 tender-items and bracket it with four estimators (OLS, cross-fit, CEM, IPW). It is descriptive corroboration that the screening signal concentrates on items that also carry a price imprint, not a causal target. We also report the overlap-restricted association β^{ov} (Table 10), and the decomposition in §7.2 shows that the broad-sample positive imprint survives an unweighted overlap restriction at +0.044 and reverses to −9.72% only under ATT-reweighting that up-weights cells with relatively few untreated counterfactuals.

Two design-based strategies that return null. A sharp regression discontinuity at the convite/pregão statutory value caps is gated by the absence of bunching: McCrary-style density tests around the R\$80,000 cap return a left/right ratio of 0.94 and a log-density discontinuity of −0.063 (Table B.5), with the convite share dropping cleanly from 69% below to 59% above. Local first-stage power is too weak to support a fuzzy RDD at standard bandwidths. A difference-in-differences design using Decreto 9.412/2018's 2018 raise of the convite cap fails parallel trends at the available pre-treatment window. We report both failures openly. They do not constrain the screening interpretation, which does not require a causal identification of β, but they do constrain alternative framings of the empirical object that would.

5.2. Reduced-Form Specification

The reduced-form specification is

$$y_{igt} = \beta \cdot \text{losers}_{igt} + \mathbf{x}'_{igt}\boldsymbol{\delta} + \alpha_g + \lambda_t + \gamma_k + \varepsilon_{igt} \tag{1}$$

where y_{igt} is the outcome for tender-item i in item group g at time t and procuring unit k; losers_{igt} indicates the presence of at least one frequent-loser participant; $\mathbf{x}_{igt}$ includes a convite indicator and any specification-specific controls; α_g, λ_t, and γ_k are item, year, and procuring-unit fixed effects; and ε_{igt} is the disturbance, clustered at the item level. The headline outcome is $\log p_{\text{negotiated}}$; auxiliary outcomes (number of bidders, number of bids, number of non-frequent-loser bidders) are reported in §7.1 and the appendix.

We estimate Equation (1) under four specifications. The general within-item-and-year baseline drops the procuring-unit fixed effect; the within-procuring-unit specification adds it. Two modal subsamples, pregão-only and convite-only, decompose the headline along the observability-regime contrast set up in §3. The estimation sample comprises 1,654,401 tender-items (§4.3).

Three complementary estimators bracket Equation (1). Cross-fitting defines the frequent-loser list on odd years, estimates the equation on even years, and reverses the assignment, removing within-sample classification noise. Coarsened exact matching (CEM) and inverse-probability weighting (IPW) adjust for selection on observables (year, procedure type, item group, procuring-unit-size quartile). Under the screening framing all three estimators bracket the broad-sample object β rather than a treatment-effect target; the four-estimator bracket gives the conditional range $+3.6\% - +7.7\%$ used throughout the paper (§7.1). A leave-one-out instrumental-variable specification uses the supply of frequent losers in other procuring units in the same product market and year as instrument; the IV coefficient is informative as a direction-of-attenuation diagnostic, returning a magnitude larger than OLS, the direction predicted when the binary indicator coarsens the underlying continuous loss-intensity primitive (§8.1). We do not interpret the IV estimate as a causal magnitude.

Standard errors are clustered at the item level as the baseline choice. Robustness to procuring-unit clustering and to two-way Cameron et al. (2011) clustering is reported in §9.1.

5.3. Identification Audits as Screening-Value Diagnostics

The screening framing reorganizes the role of the four identification audits reported in §9.2. Each audit answers a distinct question about the screening object rather than a question about the credibility of a causal estimate.

The robustness values of Cinelli and Hazlett (2020) ($RV_{q=1} = 17.5\%$) and Oster (2019) ($\hat{\delta} = 261.64$) bound the broad-sample association against selection on observables already absorbed by item, year, and procuring-unit fixed effects. They calibrate how much unobserved-confounder variance would have to load on observables to explain β away. Under the screening framing this is the bound on selection-on-observables explanations of the screen's signal.

Strict-overlap matching delivers the second empirical object β^{ov} (Table 10). Its role is not to sharpen β but to instantiate the alternative empirical object whose contrast with β identifies the screening-value interpretation.

The leakage audit (Table 7) decomposes the in-sample item-level discrimination AUC into a structural component and a tautology component, by holding cobidder firms out of score construction within cross-validation folds and by evaluating under temporal holdout. It is a generalization diagnostic for the screening statistic, not an identification test for β.

The sub-threshold-always-loser placebo (Table C.2) confirms that the participation primitive operates through the threshold-crossing population the construct identifies, rather than through generic always-loser supply. It is a specification diagnostic for the participation primitive, not an identification test for β.

The four audits are reported in §9.2 and the identification scope they delineate is consolidated in §11.1.

5.4. Heterogeneity Across Detection Regimes

The framework of Online Appendix A maps procuring-unit size to the principal detection cost θ_k through the comparative static $\partial m^*/\partial\theta_k < 0$. We split the sample into quartiles of procuring-unit size (annual contract volume in BEC) and re-estimate Equation (1) within each quartile, with robustness to alternative size measures (cumulative item count, headcount of distinct purchasers). Under the screening framing the gradient reads the screening signal's informativeness across detection regimes (§7.3); we do not interpret it as identification of an oversight channel. A second descriptive heterogeneity specification, in winner-concentration and repeated-pair density at the frequent-loser-firm level, is reported in §11.2 as scope of the screening object rather than as a positive test of cartel-rotation predictions.

Summary. The empirical strategy is organized around the discrimination object first and the conditional association second. The validation in §6 carries the discrimination evidence against three cartel-adjacency targets and the within- stratum theory-validation bridge to the cover-bidder type. The within-item price association in §7 reports the broad-sample β, its overlap-restricted counterpart β^{ov}, and the empirical decomposition of the sign reversal as a reweighting phenomenon rather than an overlap-dropping phenomenon. The architectural test in §10 reads the screening statistic against bid- distribution detectors trained on bid microdata. None of the defended claims relies on a causal identification of β, and the failed RDD/DiD designs reported above do not constrain any of them.

6. Validation: Cartel Adjacency Under Coarsened Observability

The validation answers two questions in sequence. First: does the screening statistic concentrate on populations with a higher prior probability of operating inside cartel-affected environments than random matching at comparable participation volume? Second: does the concentration appear in evidence that predates estimation, so that the result is not an artifact of the within-sample CADE record? We answer both in the affirmative, against the structurally appropriate validation object given the data layer.

That object is cartel *adjacency*, not membership. The award-layer envelope (§3) does not contain the information required for firm-level membership identification, and the screening framing of §5.1 does not require it: the participation primitive the framework identifies is precisely a loser-side exposure measure, not a winner-side label. The cobidder population—always-loser firms that participated alongside CADE direct defendants—is therefore the empirical content the screening statistic is designed to recover. We report against direct-defendant labels separately to bound the construct's scope and to document the design's empirical signature.

6.1. Validation populations

The validation reads against five nested populations, each playing a distinct role in the empirical exercise. Table 4 catalogs them, with sample sizes and the function of each in the validation. The leftmost column moves from the broadest universe (all BEC participants) inward through the always-loser stratum to the flagged frequent-loser subset; the two rightmost rows split the CADE adjudication record into direct defendants—typically frequent winners and therefore outside the construct's target—and cobidders inside the always-loser stratum, which are the cartel-adjacency labels the screening statistic is designed to recover.

Table 4: Validation populations and object discipline

Population	Size	Definition and role
All BEC participants	41,444	Universe of firms bidding in BEC, 2009–2019.
Always-losers	16,843	Firms with zero wins across the sample; stratum within which the screening statistic operates.
Frequent losers	2,735	Always-losers above the median + 1.5×IQR participation threshold; the flagged population.
CADE direct defendants in BEC	47	Firms named in CADE rulings; typically frequent winners; outside the screening statistic's target population.
Cobidders in the always-loser stratum	193	Always-losers that participated alongside CADE direct defendants; the cartel-adjacency label the screening statistic is designed to recover.

Notes: The screening statistic operates within the always-loser stratum, not across the broader BEC universe. Participation alongside an adjudicated cartel member is consistent with cover bidding, capacity-transfer motives, market intelligence, or bystander participation in cartel-affected markets. Strong AUC against this label corroborates that the screening statistic concentrates on cartel-relevant environments; it does not adjudicate membership.

6.2. Within-stratum profile of cobidders

The validation positive class is cobidders inside the always-loser stratum, not direct CADE defendants. The reading would be unconvincing if cobidders were just always-loser firms with a proximity label glued on. Table 5 compares cobidders against three reference populations along four operational predictions of the cover-bidder type that the framework of Online Appendix A formalizes.

Table 5: Cobidders vs. Reference Populations: Operational Predictions of the Cover-Bidder Type

	Cobidders ($N = 191$)	FL non-cobidders	Always-loser, non-FL	Other winners	Cohen's d vs. FL	Wilcoxon p
Mean portfolio item-group HHI	0.380	0.288	0.719	0.294	+0.39	< 0.001
Mean share of loss-bids facing a direct CADE defendant	0.015	0.002	0.002	0.003	+0.46	< 0.001
Mean share of loss-bids in winner-pairs of count ≥ 5	0.216	0.334	0.026	0.412	−0.38	< 0.001
Mean total participations	136.471	76.652	12.119	1603.320	+0.67	< 0.001

Notes: Each row reports a firm-level mean for four populations: cobidders (193 always-losers participating alongside defendants), FL non-cobidders (2,735–193 frequent losers without recorded direct-defendant adjacency), always-losers threshold, and other winners (firms with at least one win, excluding direct CADE defendants). The two rightmost c Cohen's d and a Wilcoxon rank-sum test of cobidders against FL non-cobidders, the relevant within-stratum compari operational measures map directly to the cover-bidder type in Online Appendix A: total participation (deployment int of own loss-bids facing a direct CADE defendant (deployment proximity), share of loss-bids in winner-pairs of coun co-bidding density), and portfolio item-group HHI (specialization in the cartel's market).
Source: `scripts/60_theory_validation_bridge.R`.

The within-stratum comparison—cobidders versus FL non-cobidders, both above the FL participation threshold—moves in the direction the framework predicts on 4 of the 5 measured dimensions where the data layer can speak. Cobidders deploy at higher intensity: 136.5 mean total participations against 76.7 for FL non-cobidders, Cohen's $d = +0.67$, $p < 10^{-15}$; the gap is sharper still on unique winners faced (24.8 vs 13.5, $d = +1.00$, $p < 10^{-22}$). They are deployed proximally to the cartel: 1.5% of their loss-bids face a direct CADE defendant against 0.2% for FL non-cobidders, $d = +0.46$, $p < 10^{-70}$. They specialize in narrower market slices: portfolio item-group HHI of 0.380 against 0.288 ($d = +0.39$, $p < 10^{-13}$), touching 7.6 distinct item-groups on average against 9.5 ($d = -0.32$, $p < 10^{-6}$). The one dimension that moves the wrong way is the share of own loss-bids in repeat-pair clusters of count ≥ 5 (21.6% vs 33.4%, $d = -0.38$); we note it without leaning on it.

The contrast with non-FL always-losers is sharper still. Firms below the FL threshold participate in approximately 12 tenders each, face 2 unique winners on average, touch only 2 item-groups, and almost never face a direct CADE defendant: a thin-participation, narrow-portfolio profile that does not look like the cover-bidder type. The cobidder population is empirically distinct from this reference group as well as from FL non-cobidders.

The bridge is partial. The framework also predicts firm-level bid aggressiveness and bid dispersion patterns; the award-record data layer does not preserve per-firm bid amounts and these predictions cannot be tested directly here. This is a portability point: in jurisdictions where bid microdata is preserved at the firm level, the bridge could be extended in those two further dimensions (§11, §10). Within what the data can measure, cobidders match the modeled type along four of five operational predictions, including the two with the largest standardized differences. We read the discrimination evidence in the rest of this section against an empirically grounded validation object, not against a label of convenience.

6.3. Conservative benchmark

The primary benchmark restricts the ground truth to 4 CADE cases adjudicated before 2020 (coleta de lixo, aquecedores solares, trens-metrôs, tecnologia-informação 2019-10): 30 firm-defendants, all active in BEC, with 210 always-losers co-bidding with at least one of them, of which 108 are frequent losers.

The relevant comparison is whether frequent losers co-bid with direct defendants at rates above what random matching at comparable participation volume would produce. We construct the counterfactual baseline by stratified permutation: for each of 1,000 iterations, we permute the cobidder/non-cobidder labels within strata defined by participation-volume quartile, re-compute the share of frequent losers among the relabeled cobidders, and accumulate the empirical distribution under the null of random matching conditional on participation volume. The participation-stratified co-bidding rate observed in the data is 3.95% against a permutation baseline of 1.24% ($p < 0.001$, 1,000 iterations), an excess ratio of 3.2×. The ROC AUC against the contemporaneous ground truth is 0.748 (95% DeLong CI $[0.713, 0.783]$).

This is the primary benchmark for the screening interpretation on three grounds. The construct uses no information from cases adjudicated post-sample. The labels predate estimation: cartel existence and membership were established by enforcement before the screening statistic was built. And the permutation baseline strips out the mechanical co-bidding rate that participation volume would produce by itself, isolating the residual concentration the screening statistic is meant to identify. The 3.2× excess is what the screening framing predicts under the separating equilibrium of Online Appendix A: where the cartel deploys cover bidders, the loser-side participation footprint concentrates above the random-matching baseline.

6.4. Prospective benchmark and within-firm enrichment

A more permissive benchmark uses CADE's full procurement-cartel portfolio (12 cases, 65 firm-defendants, of which 47 are active in BEC). Among 2,735 frequent losers, 193 (7.1%) co-participate with at least one of the 47; the participation-stratified excess ratio is 3.5× ($p < 0.001$). Firm-level AUC is 0.924 (95% CI [0.921, 0.926]) in-sample and 0.864 ([0.858, 0.870]) under temporal holdout (train 2009–2016, test 2017–2019). The benchmark treats post-sample adjudications as informative labels for cartel involvement during the BEC window—a reasonable but non-trivial assumption that the conservative benchmark in §6.3 avoids. We report it as complementary, not as the primary anchor.

Figure 1 reports the receiver-operating characteristic curves under the rolling-origin temporal-holdout scheme, with one curve per test year and the diagonal reference. The curves separate cleanly from the diagonal across all six test years; the year-by-year AUC is reported in Appendix C.3.

Permutation null and leakage decomposition.. Two further diagnostics tighten the discrimination evidence. The participation-stratified permutation null behind the conservative benchmark is reported in Table 6: under 1,000 iterations of within-quartile relabeling of cobidder/non-cobidder status, the empirical excess ratio of 3.2× falls in the upper tail of the null distribution at $p < 0.001$. The leakage decomposition in Table 7 separates the in-sample item-level discrimination AUC of 0.939 into a structural component recovered when cobidder firms are held out of score construction within cross-validation folds (AUC $\approx$ 0.86–0.89) and a mechanical-tautology component ($\approx$ 0.10–0.13). The structural component is what the screening interpretation rests on.

A within-firm exercise on the same portfolio: 3 of 7 always-loser direct defendants (43%) cross the frequent-loser threshold against a population baseline of 16%. Table 8 lists them.

Table 6: CADE Validation: Co-Participation and Permutation Test

Panel A: Co-participation with CADE Cartelists		
CADE procurement cartel cases (unique processes)	12	
firm-defendants across all cases	65	
final adjudication dates	2015–2025	
cases adjudicated after 2019 (prospective)	8	
CADE firm-defendants matched to BEC via CNPJ	47	
	FL Firms	Non-FL Firms
Co-bids with CADE	193	2,290
Does not co-bid	2,542	36,419
Total	2,735	38,709
Co-participation rate	7.1%	5.9%
χ^2 statistic	5.7	
p-value	0.017	
Odds ratio	1.21	
Panel B: Permutation Test (1,000 iterations)		
Observed FL-CADE rate	0.0706	
Permutation mean rate	0.0201	
Permutation SD	0.0026	
p-value (rate $\geq$ observed)	0.0000	

Notes: Panel A: contingency table of firm co-participation with CADE cartel-defendants. Eight of the 12 cases were adjudicated after the BEC sample ends (post-2019), so the validation is partially prospective: it asks whether 2009–2019 FL flags anticipate firms later confirmed by enforcement. Panel B: 1,000 random draws of 2,735 firms from the always-loser pool, stratified by participation quartile, computing co-participation rate with CADE firm-defendants.

Table 7: Leakage Audit of the Item-Level Loser-Side Concentration AUC

Audit specification	What it tests	AUC	95% CI
Original (in-sample, full pool)	Reference: log(1+max tc) predicting any-cobidder participation.	0.995	$[0.995, 0.995]$
Audit 1: Different label, same score (scope)			
Same score $\rightarrow$ any-direct-CADE label	Whether the screen also flags items with direct CADE defendants ($N+$=54,039 items).	0.506	$[0.505, 0.507]$
Audit 2: 5-fold cross-validation at the cobidder-firm level (tautology)			
Pooled out-of-fold	Hold out $\sim 1/5$ of cobidders per fold; recompute tenders_count without their participation; predict items containing held-out cobidders using the recomputed score. Pool out-of-fold predictions across folds.	0.891	$[0.887, 0.894]$
Audit 3: Temporal holdout (generalization)			
Train 2009–2016 $\rightarrow$ test 2017–2019, any-cobidder	Compute tenders_count using only 2009–2016 items; predict items in 2017–2019.	0.864	$[0.859, 0.870]$
Train 2009–2016 $\rightarrow$ test 2017–2019, any-direct-CADE	Same temporal split, against direct-CADE label.	0.511	$[0.510, 0.513]$

Notes: The in-sample AUC of 0.995 is partly tautological by construction. Audit 2 isolates the structural component by holding cobidder firms out of score construction within each fold; Audit 3 isolates it temporally by computing the score on 2009–2016 participation only. AUC retains ≥ 0.85 under both audits; the 0.10–0.13 drop is the pure-leakage component. Audits against direct-CADE labels return random AUC, consistent with the scope asymmetry in Table 8.

Table 8: Direct CADE defendants flagged by the construct

Firm	Cartel	CADE process	Tenders	Wins
Sol Tecnologia	Solar water heaters	08012.001273/2010-24	84	0
Nova Esperança Locadora	School transportation	08700.005876/2019-85	97	0
Jofran Comércio	Trash bags (Op. Colludium)	08700.005789/2015-02	65	0

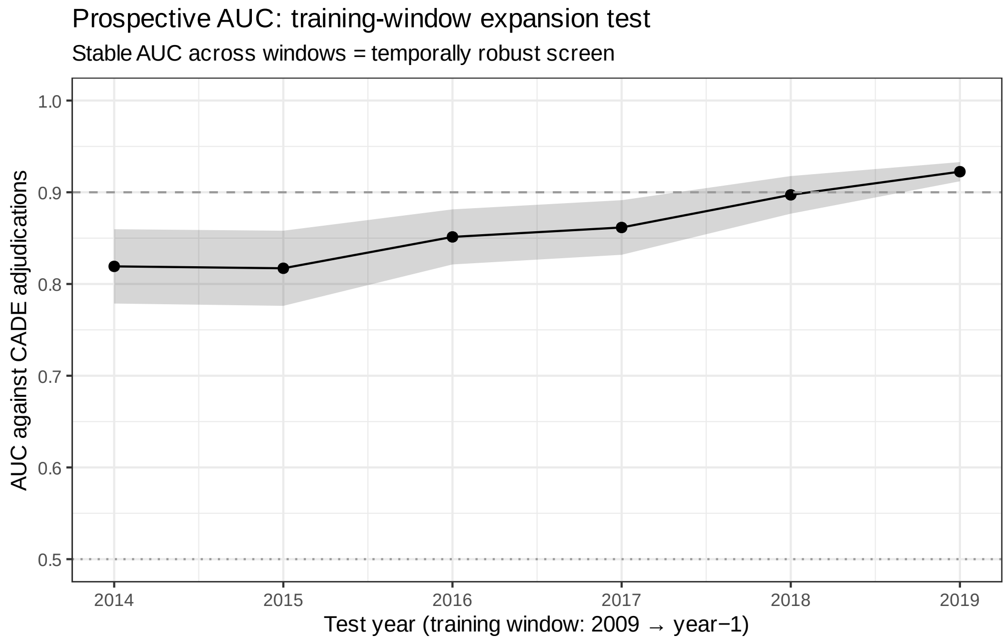

Figure 1: ROC curves for the construct under rolling-origin temporal holdout. Training window is 2009 through test year − 1; the test year provides the out-of-sample cobidder ground truth. AUC values, 95% CIs, sample sizes, and CADE counts per year are reported in Table C.3. Source: `scripts/17_temporal_holdout_roc.R`.

These are the exception within a direct-defendant population that is otherwise frequent-winner-heavy; the AUC against direct defendants in the broader BEC firm universe remains 0.49. The asymmetry between cobidder discrimination (0.924 in-sample, 0.864 under temporal holdout) and direct-defendant discrimination (0.49) is the design's empirical signature, not a failure of validation. The screening statistic recovers loser-side participation footprints; direct defendants are by construction the winner side of the same arrangements and lie outside the target population.

6.5. Robustness to the enforcement record

Dropping all 31,447 CADE-involved tender-items and re-estimating the within-PBU baseline yields $\hat{\beta} = 0.062$ ($N =$ 1,622,954), virtually unchanged from 6.4%. The broad-sample screening signal therefore does not depend on the within-sample CADE adjudications for its content: removing the items where the cartel record bites leaves the conditional association intact. Under the screening framing this is the predicted outcome—the participation footprint the construct flags is a loser-side equilibrium object, present wherever the deployment problem operates, and not a feature mechanically produced by the items already labeled by enforcement. The robustness completes the conservative benchmark of §6.3: the labels that drive the validation are stripped out of the estimation sample, and the price-imprint signal survives.

6.6. What the validation does and does not show

The validation corroborates the screening interpretation along three dimensions, each consistent with the framework of Online Appendix A. The participation-stratified excess of 3.2× on the conservative benchmark, the discrimination AUC of 0.748 on cases adjudicated before the sample window closes, and the 3.5× excess on the full prospective portfolio (AUC 0.924 in-sample, 0.864 under temporal holdout) all read in the same direction: within the always-loser stratum, frequent losers concentrate near adjudicated cartel members at rates well above participation-stratified random matching.

Three boundaries delimit what the validation supports. The labels are indirect by construction, a feature of working at the award layer rather than the bid layer; the AUC asymmetry between cobidders and direct defendants (0.924 versus 0.49) is the empirical signature of a screening statistic that recovers loser-side participation, not winner-side identity; and the screening signal in β persists when CADE-involved items are removed from estimation, which means the validation labels are not the mechanism that produces the price-imprint result.

The validation establishes that the screening statistic flags cartel-adjacent environments. Under coarsened observability it cannot adjudicate firm-level membership, and the screening framing does not require it to.

7. Main Results

This section delivers the screening-value evidence in three moves. First, the broad-sample association β across four estimators (§7.1). Second, the overlap-restricted association β^{ov} and the sign reversal as the diagnostic that locates the construct as a screening-value object rather than a treatment-effect object (§7.2). Third, the heterogeneity of the screening signal across detection regimes (§7.3).

7.1. Broad-Sample Conditional Association (β)

Table 9 reports OLS estimates of Equation (1). Frequent-loser presence associates with a higher log negotiated unit price in all four specifications: 6.8% within item and year (6.4% adding procuring-unit fixed effects), and 9.3% in pregão against 3.8% in convite. The comparison set is the 1,654,401 tender-items in the same product-code × year × procuring-unit cells without frequent-loser participants.

Table 9: Negotiated Prices (log): Effect of Frequent Losers

	(1) General	(2) General	(3) Pregão	(4) Convite
FL presence	0.0677***	0.0636***	0.0933***	0.0382**
	(0.0230)	(0.0215)	(0.0255)	(0.0186)
Convite	−0.0090**	0.0095***		
	(0.0037)	(0.0035)		
Observations	1,654,401	1,654,401	546,549	1,107,852
R^2	0.879	0.886	0.882	0.893
Item FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
PBU FE	NO	YES	YES	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Columns (3) and (4) restrict the sample to pregão and convite procedures.

The pregão–convite contrast is informative on whether the construct's signal could be mechanically driven by the convite minimum-bidder rule (Lei 8.666/93, Art. 22, § 3): a quorum-filler reading would predict a sharper imprint in convite, but the data show the opposite (+9.59% in pregão vs. +3.92% in convite; Table B.4). The signal is therefore not reducible to the quorum rule. We read the asymmetry as informativeness across observability regimes (§8.2), not as a positive test of any specific institutional channel.

Cross-fit estimation yields 3.6%, CEM +0.077, and IPW 0.055. A leave-one-out IV returns 0.194 ($p < 0.05$, first-stage $F = 396$), the direction expected when the binary indicator coarsens the underlying continuous loss-intensity statistic (§8.1). The four broad-sample estimators jointly bound the conditional range $+3.6\% - +7.7\%$.

Under the screening framing of §5.1, β is the empirical content the construct is designed to recover: the conditional association at items where frequent losers deploy, integrated over the cartel's deployment distribution. The relevant question for §7.2 is whether the deployment sorting is itself part of the screening-value content, not whether β would survive a counterfactual that strips the sorting away.

7.2. Overlap-Restricted Association (β^{ov}) and What the Sign Reversal Does and Does Not Tell Us

Restricting the comparison to cells in which frequent-loser-present and -absent items genuinely overlap on observables and reweighting to the average treatment effect on the treated produces a reversed conditional coefficient: overlap-cell ATT −9.72%, propensity-score-trimmed ATT −30.67% (Table 10).

Table 10: Item-Level Scope Checks: Conditional Association Under Stricter Overlap

Specification	Coefficient	SE	p-value	N
baseline_fe	+0.064	0.021	0.003	1,653,658
overlap_cell_att	−0.097	0.015	< 0.001	1,517,066
overlap_ref_att	−0.097	0.015	< 0.001	1,434,636
ps_att_trimmed	−0.307	0.020	< 0.001	400,687

Notes: Reports β (`baseline_fe`) and the overlap-restricted β^{ov} (`overlap_cell_att`, `overlap_ref_att`, `ps_att_trimmed`). Under the screening framing of §5.1 these are different empirical objects rather than alternative estimates of the same parameter; the sign reversal is the diagnostic that locates the construct as a screening-value object.

The reversal is real, and we report it. Two interpretive moves, however, that an earlier version of the paper made are not supported once the gap is decomposed empirically. We document the decomposition here so that the reader can see exactly what the data do and do not say.

The reversal is a reweighting result, not an overlap-dropping result. The gap from +0.064 (1,654,401 items) to −0.097 separates into two margins: (i) restricting to cells with both treated and untreated items, and (ii) ATT-reweighting within those cells. We re-estimated the same regression on the overlap subsample *without* ATT-weights. The unweighted overlap coefficient is $+0.044$ ($p = 0.035$, $N = 1{,}517{,}868$). The reversal to -0.097 is produced by the ATT-weighting step, not by the overlap restriction itself: only 839 of 79,452 treated items (1.06%) lack a within-cell counterfactual and are strictly dropped. The remaining 99% of treated items participate in both estimates; the difference is which untreated counterfactuals each estimator up-weights.

The cell-dropping pattern provides only mixed support for a deployment-sorting reading. An earlier version of the paper read the reversal as a diagnostic that overlap restriction strips out exactly the deployment-sorted, screening-rich environments. Comparing treated items in dropped versus surviving cells along six discriminating dimensions (Table 11, Panel C): convite share is *lower* in dropped cells (0.496 vs. 0.638, $p < 10^{-15}$, opposite of the deployment-targeting prediction); procuring-unit-size quartile is also lower in dropped cells; direct-CADE-item share differs trivially and is statistically null (0.019 vs. 0.024, $p = 0.33$); cobidder-item share is the one dimension *consistent* with the deployment reading (0.139 vs. 0.078, $p < 10^{-6}$). Dropped cells are dominated by high-priced specialty items with many bidders—a thin-market problem distinct from any deployment-sorting story. We no longer claim that the cell-dropping pattern confirms the screening-value reading; the empirical evidence is mixed at best.

What survives inside the overlap sample. Two results inside the overlap subsample remain informative for the framework. First, $\widehat{\beta}^{\text{ov}}$ is statistically null among items containing direct CADE defendants (-0.061, $p = 0.45$), in contrast to its strongly negative value in non-direct-CADE items (-0.097, $p < 0.001$): on the cells closest to adjudicated cartel activity, the negative reading does not survive. Second, in the highest tender-value quartile (Q4)—where deployment value is most plausible *a priori* on contract size alone—$\widehat{\beta}^{\text{ov}}$ is positive and significant at $+0.041$ ($p = 0.045$). The two quantities are modest but they are residue, inside the overlap sample, of what the screening framework would predict; we report them as such, not as positive identification.

What we read into the reversal, and what we do not. Under the framework of Online Appendix A, β on the broad sample picks up the conditional association at items where

Table 11: Sign-Reversal Decomposition: Where Does $\widehat{\beta}^{ov} < 0$ Live, and Which Items Get Dropped?

	Coef. / Mean	SE	p-value	N
Panel A. Headline specifications on common sample.				
Broad sample $\widehat{\beta}$	+0.064	0.021	0.003	1,654,401
Overlap-cell ATT $\widehat{\beta}^{ov}$	−0.097	0.015	< 0.001	1,517,868
Panel B. $\widehat{\beta}^{ov}$ within overlap-cell sample, by subgroup.				
$\widehat{\beta}^{ov}$ in non-direct-CADE items	−0.097	0.016	< 0.001	1,487,055
$\widehat{\beta}^{ov}$ in direct-CADE items	−0.061	0.080	0.445	30,813
$\widehat{\beta}^{ov}$ in pregão items	−0.099	0.020	< 0.001	475,923
$\widehat{\beta}^{ov}$ in convite items	−0.098	0.013	< 0.001	1,041,945
Panel C. Treated items in dropped vs. surviving cells (means).				
	In dropped	In surviving	Difference	t-test p
Convite share among treated items	0.496	0.638	−0.142	< 0.001
Direct-CADE-item share among treated items	0.019	0.024	−0.005	0.333
Cobidder-item share among treated items	0.139	0.078	+0.061	< 0.001
Mean log reference price among treated items	6.691	5.490	+1.201	< 0.001
Mean number of bidders among treated items	10.533	9.047	+1.486	< 0.001

Notes: Panel A reports the broad-sample $\widehat{\beta}$ and the overlap-cell ATT-weighted $\widehat{\beta}^{ov}$, both estimated on items with non-missing log negotiated price and within-item, year, and PBU fixed effects. Panel B re-estimates $\widehat{\beta}^{ov}$ on the same overlap-cell ATT design within four subgroups (pregão vs. convite items; non-direct-CADE vs. direct-CADE items). Panel C compares treated items (FL present) that fall in cells dropped by the overlap restriction (no untreated counterfactual) versus treated items in surviving cells; differences in convite share, direct-CADE-item share, cobidder-item share, log reference price and number of bidders test the screening-value reading directly. The screening framework predicts that items dropped by overlap restriction should disproportionately carry markers of deployment value (higher convite, direct-CADE, cobidder-item rates).
Source: `scripts/59_sign_reversal_decomp.R`.

frequent losers deploy, integrated over the cartel's deployment distribution. β^{ov} on the ATT-reweighted overlap sample is a different empirical object that up-weights cells with relatively few untreated counterfactuals, and the reweighting can generate a negative coefficient under data-generating processes the framework allows (Online Appendix A, Proposition 4, treated as one interpretation among several). What we now claim is narrower than v15 of this paper claimed: (i) the broad-sample positive imprint *survives* an unweighted overlap restriction at +0.044, so it is not built on items the design would discard; (ii) the cell-dropping pattern offers one of six empirical dimensions consistent with deployment sorting (cobidder concentration); (iii) inside the overlap sample, $\widehat{\beta}^{\text{ov}}$ is null on direct-CADE items and positive in the largest tender-value quartile. What we do *not* claim is that the sign reversal cleanly identifies the screening-value reading against alternative data-generating processes. The pairing of β and β^{ov} is reported transparently; we privilege β under the screening framing and acknowledge that β^{ov} would be privileged under a treatment-effect framing the data do not adjudicate.

The construct's defense, in the rebuilt hierarchy, does not rest on the sign reversal. The supporting evidence carries: the discrimination AUCs and the architectural comparison with bid- distribution detectors (§6, §10) do not depend on overlap weighting. Cinelli and Hazlett (2020)'s $RV_{q=1} = 17.5\%$ and Oster (2019)'s $\hat{\delta} = 261.6$ bound the broad-sample association against selection on observables already absorbed by the fixed effects.

Figure 2 reports the same coefficient trajectory across the four anchor specifications.

7.3. Heterogeneity Across Detection Regimes

Splitting the sample into quartiles of procuring-unit size and re-estimating the within-item specification produces a sharp contrast at the extreme quartiles: +21.4% at the smallest-buyer quartile (Q1) and +1.7% at the largest (Q4), a 12.6× extreme-quartile gradient. The two intermediate quartiles—Q2 at +9.8% and Q3 at +4.5%—are imprecisely estimated and not statistically distinguishable from each other or from Q4 (item-clustered standard errors; Table B.3). We do not claim a clean monotonic decline across all four quartiles; we claim that the extremes contrast in the direction the framework predicts and that the contrast is preserved across alternative size measures (annual contract volume,

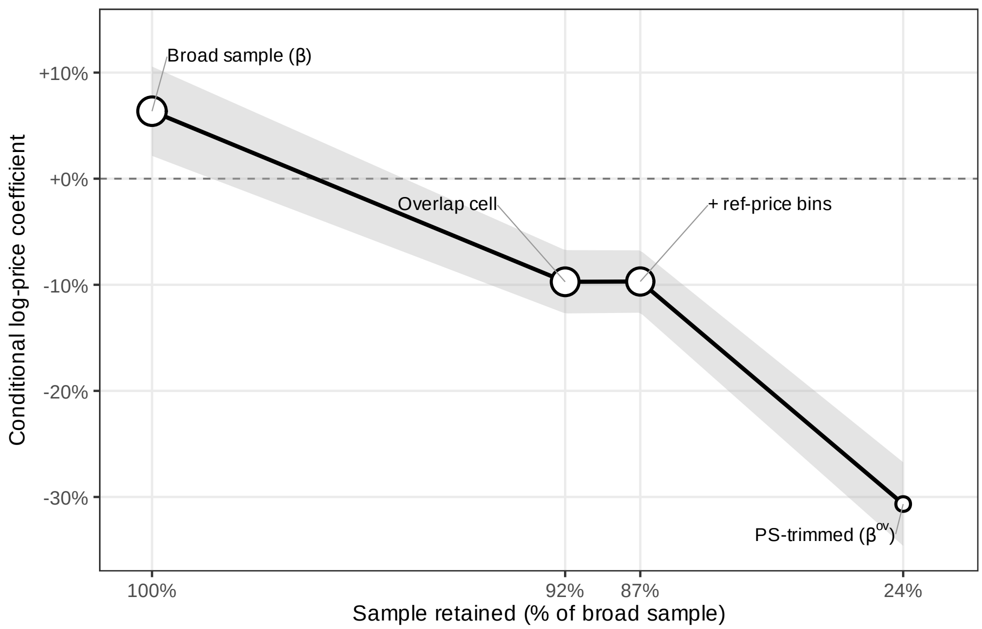

Figure 2: Conditional log-price coefficient across the four specifications of Table 10. Point size encodes sample retention. The shaded band is the 95% confidence interval. The figure displays the gap between β and β^{ov}; §7.2 decomposes the gap into an overlap-restriction step (small) and an ATT-reweighting step (the bulk).

cumulative item count, headcount of distinct purchasers).

The framework in Online Appendix A predicts a contrast in this direction through the detection-cost comparative static $\partial m^*/\partial\theta_k < 0$: cover-bidder deployment is more aggressive where the principal cost of detection is lower, so the screening signal is more pronounced in the small-buyer regime. We read the Q1-versus-Q4 contrast as heterogeneity in the screening object across the monitoring environment, not as identification of a causal channel. Buyer size proxies for several correlated institutional features—procurement-officer tenure (Coviello and Gagliarducci, 2017), internal-audit infrastructure, item composition, and discretionary procedure use (Decarolis et al., 2020; Decarolis and Giuffrida, 2017)—none of which can be separated with the variation we have. The implication is that the screening signal varies systematically with features of the enforcement environment, in the direction the framework's deployment problem predicts, with the magnitude of the variation concentrated at the extremes of the buyer-size distribution.

Summary. The three results read jointly. The broad-sample β picks up the participation footprint where the cartel deploys; the overlap-restricted β^{ov} isolates the within-support comparison; the extreme-quartile contrast locates the regime where the screening signal is sharpest. The directions match the framework's predictions (broad-sample positive, overlap-restricted negative, small-buyer larger than large-buyer), and the validation in §6 corroborates the reading on an independent dimension. What remains is the modal contrast and the predictions the data do not adjudicate (§8).

8. Heterogeneity of the Screening Signal

The screening object is read against the framework's deployment problem. Two pieces of within-evidence discipline the framing. First, the continuous loss-intensity statistic strictly dominates the binary rule in discrimination, consistent with Proposition 2's identification of $\log(1 + \text{tenders\_count})$ as the sufficient ranking statistic (§8.1). Second, the modal contrast between pregão and convite reads as variation in the observability regime within which the screening signal is recovered (§8.2). Two further framework predictions—a cell-heterogeneity grid and a discrete first-time-FL imprint—are reported in §11.2 and Online Appendix C; the data do not adjudicate them under disciplined estimation, and the screening interpretation does not require them.

8.1. Loss Intensity Dominates the Threshold Cutoff

The binary rule that defines frequent losers (more than 13.5 tender-items, zero wins) is the information-coarsening of the participation primitive identified by the framework, not a primitive in itself. Proposition 2 identifies $\log(1 + \text{tenders\_count})$ as a sufficient ranking statistic for cover-bidder type given award-record data; the binary rule is the information-coarsening of the latent posterior under the separating equilibrium of Online Appendix A. The horse race in Table 12 confirms the prediction empirically.

On a harmonized same-sample basis, the continuous $\log(1 + \text{tenders\_count})$ score discriminates the 193 adjudicated CADE-cobidder firms inside the always-loser stratum at AUC 0.939 (95% CI [0.932, 0.946]), against AUC 0.911 ([0.898, 0.925]) for the binary FL14 indicator (DeLong et al. 1988 test, $Z = -4.30$, $p = 2e - 05$). The continuous

Table 12: Horse Race: FL14 Binary vs Continuous Loss Intensity (Same Sample)

Specification	Coef on FL14	Coef on log(1+ max tc)	Convite	N
(1) FL14 only	+0.0653***	–	+0.171***	1,653,658
	(0.0216)		(0.013)	
(2) Continuous only	–	+0.0188***	+0.169***	1,653,658
		(0.0055)	(0.013)	
(3) Both, additive	−0.0746*	+0.0349***	+0.169***	1,653,658
	(0.0383)	(0.0108)	(0.013)	

Firm-level AUC against 193 *CADE-cobidder labels (always-loser pool,* $N = 16{,}843$*)*

FL14 binary	AUC = 0.911 [0.898, 0.925]
log(1+tenders_count)	AUC = 0.939 [0.932, 0.946]
DeLong test (continuous vs FL14)	$Z = -4.30$, $p = 2e-05$

Notes: Outcome is $\log p_{\text{negotiated}}$ within the v13 panel of 1,653,658 items. Item-level treatment is **1**[any FL14 participant] in column (1) and the log of the maximum loss-intensity score among always-loser participants in column (2). Column (3) includes both. The FL14 coefficient flips sign when continuous intensity is added: the binary discretization at 14 tenders is the information-coarsening of the underlying continuous signal, not an economically primitive threshold. SEs clustered at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

statistic is strictly more informative: the ∼2.8-point AUC gain on the same firm panel is the dose-response signature the framework predicts when the binary rule coarsens a continuous primitive.

Standalone price coefficients align with the discrimination result: FL14 binary +0.0653 ($p < 0.001$), continuous $\log(1 + \max \text{tc})$ +0.0188 ($p < 0.001$); Table 12, columns 1–2. When both are included additively (column 3), the FL14 coefficient flips sign (−0.0746, $p < 0.05$) while the continuous coefficient nearly doubles (+0.0349, $p < 0.001$). The flip is the framework's prediction empirically: once the continuous primitive is in the specification, the binary indicator captures only tail residuals against the continuous dosage, and reading the binary in isolation overstates the cover-bidder type information by absorbing the variation the continuous primitive carries. The dose-response pattern is therefore consistent with the screening interpretation and inconsistent with a pure-confound reading of the broad-sample association (§7.2).

We use the binary FL14 rule as the discrete cut-off and the continuous statistic as the discriminating instrument. The headline range $+3.6\% - +7.7\%$ in §7.1 is computed on the binary; the AUC headline 0.924 (95% CI [0.921, 0.926]) reported in §6.4 is binary firm-level discrimination.

8.2. *Modal Contrast as Observability-Regime Variation*

The screening signal is sharper in pregão than in convite along two independent dimensions of evidence. On the price-imprint side, the within-modal estimates of §7.1 place the binary specification at +9.59% ($p < 10^{-3}$) inside pregão versus +3.92% ($p = 0.037$) inside convite, a ratio of 2.45× on the larger pregão sample. On the discrimination side, the modal-by-modal AUC against the 193 adjudicated CADE-cobidder firms is 0.952 in pregão primary auctions versus 0.816 in convite primary auctions, with a bootstrap difference of -0.136 ($p \approx 0$). The two dimensions point in the same direction; we report and interpret both with attention to a sample-size caveat the convite-modal AUC carries.

Sample-size caveat on the convite-modal AUC. Only 6 adjudicated CADE-cobidder firms fall in the within-convite stratum, against several dozen on the pregão side. Confidence intervals on the convite-modal AUC are correspondingly wider than the point estimate suggests, and we treat the convite AUC as a directional indicator rather than as a precisely identified estimate. The price-imprint evidence in §7.1, computed on the much larger 1,105,852 convite tender-items, carries the statistical weight; the convite-modal AUC corroborates the direction.

Why the data reverse the naive institutional reading. The convite minimum-bidder rule (Lei 8.666/93, Art. 22, § 3) is the obvious institutional suspect: a quorum-filler reading would predict a sharper screening signal where the rule binds, that is, in convite. The data reverse this prediction. We do not read the reversal as a positive test of any alternative institutional theory; we read it as variation in the screening signal across observability regimes (§3), with the design unable to separate two compatible mechanisms within that regime. The pregão environment exposes a real-time bidder-by-bidder interaction footprint that convite's sealed-bid envelope does not, and the bidder-pool composition that survives convite's quorum requirement may itself be more homogeneous than the pregão pool. Both mechanisms predict a sharper screening signal in pregão; the design separates neither.

The implication for portability is asymmetric. The post-sample reform (Lei 14.133/2021) consolidates pregão-style auctions as the institutional default, which is the regime in which the screening signal is sharpest. The construct travels best to jurisdictions whose observability regime resembles pregão—real-time electronic auctions with public bidder-by-bidder interaction. Award-only systems with no real-time interaction (sealed-bid invitation procedures elsewhere; some U.S. contracting-authority files) are the scope-restricted boundary, not the natural deployment environment.

9. Robustness

This section stress-tests the broad-sample association β of §7.1 and the detection-regime gradient of §7.3 against three concerns. §9.1 addresses sensitivity to the design choices embedded in the screening statistic (IQR multiplier, clustering, binary-vs-continuous specification). §9.2 reports identification audits (Cinelli and Hazlett 2020 robustness value, strict-overlap matching, leakage diagnostics, and the placebo on sub-threshold always-loser supply). §9.3 reports the temporal-holdout generalization of the screening statistic, the most consequential disclosure for the screening reading. The comparison against bid-distribution detectors is reported in the dedicated §10 on screening-vs-forensic stages.

9.1. Threshold and Specification Sensitivity

The headline price gap and its detection-regime gradient hold across reasonable variations in the design choices that define the construct. We document three: the IQR threshold multiplier, the clustering of standard errors, and the binary-versus-continuous specification of loss intensity.

The IQR multiplier governs the bright-line cutoff that separates frequent losers from the broader always-loser stratum. Our headline multiplier is 1.5, which yields a tender-count threshold of 13.5 and a flagged population of 2,735 firms. Table D.1 re-estimates the within-PBU specification under five multipliers ranging from 1.0× to 3.0× IQR. The price coefficient is positive and significant in every row, declining monotonically from +0.079 at 1.0× (threshold 10, 3,442 firms) to +0.050 at 3.0× (threshold 24, 1,456 firms). The decline is a feature, not a defect: as the cutoff tightens, the binary indicator filters fewer firms with high loss-intensity, and §8.1 established that the underlying primitive is the continuous $\log(1 + \text{tenders\_count})$ statistic. The binary at $1.5\times$ balances coverage and signal density; the construct neither evaporates at conservative thresholds nor inflates at permissive ones.

The clustering of standard errors does not change the point estimate—which is mechanical—but it does change inference. Table D.2 reports three regimes: item-level clustering (baseline, SE 0.0215), procuring-unit-level clustering (SE 0.0144, tighter than baseline), and two-way Cameron et al. (2011) clustering on item and procuring unit jointly (SE 0.0240, looser than baseline). The coefficient is significant at $p < 0.01$ under all three regimes; the baseline item-level cluster is the conservative middle choice.

The binary-versus-continuous specification of loss intensity is discussed in detail in §8.1; we summarize the implication for robustness here. The continuous $\log(1+\text{tenders\_count})$ score discriminates the same adjudicated CADE-cobidder population at AUC 0.939 (95% CI [0.932, 0.946]) versus 0.911 ([0.898, 0.925]) for the binary FL14 indicator (DeLong $Z = -4.30$, $p = 2e-05$). Standalone price coefficients in Table 12—FL14 binary +0.0653 ($p < 0.001$) and continuous +0.0188 ($p < 0.001$)—point in the same direction. The headline price gap is not an artifact of binary discretization.

9.2. Identification Audits

The construct's identification status is descriptive throughout the paper. This subsection collects four audits that calibrate the strength of the descriptive claim under specific identification concerns: omitted-variable sensitivity, common-support restriction, leakage, and a placebo on a non-cover-bidding channel.

The Cinelli and Hazlett (2020) robustness value reported in §7.1 is $RV_{q=1} = 17.5\%$: an unobserved confounder would need to explain at least 17.5% of the residual variation in both frequent-loser presence and log price to drive the coefficient to zero. The Oster (2019) bounds in Table C.1 agree directionally. The full-sample $\hat{\delta} = 261.6$ implies that unobservables would need to be more than two hundred times as important as the observables already captured by item, year, and procuring-unit fixed effects to overturn the result; the pregão subsample yields an even larger $\hat{\delta} = 634.00$. The convite subsample carries a smaller margin ($\hat{\delta} = 43.62$, $\beta^*(\delta = 1) = -0.1395$)— consistent with §8.2's finding that the construct's signal-to-noise is highest in pregão environments and that the convite estimate is the noisier modal slice. Both sensitivity tools agree: the broad-sample association is unlikely to be explained away by selection on observables of the kind already accounted for.

The strict-overlap matching exercise complements the sensitivity bounds with a different identification angle. Where Cinelli and Oster ask how strong unobserved selection would need to be to overturn the result, strict-overlap matching restricts comparisons to the cells in which frequent-loser items and non-frequent-loser items genuinely overlap on observables. Table 10 reports four specifications. The baseline within-item-and-PBU coefficient is +0.064 (n = 1,653,658, identical to the headline). Restricting to overlap cells—requiring frequent-loser and non-frequent-loser items to share the same observed pre-treatment cell—returns −0.097 (n = 1,517,066). Trimming further to common-support under propensity-score restriction returns −0.307 (n = 400,687).

The full table appears in §7.2 (Table 10). The reversal under strict overlap restricts comparisons to the items frequent losers *would* have entered with high probability, removing precisely the systematic selection differences between selected and unselected items that the broad-sample association captures (§7.1). Both signs describe the same population from different angles, neither carrying causal weight on its own; we report the four specifications jointly so the reader can locate the construct's descriptive claim within the appropriate scope.

The leakage audit in Table 7 addresses a distinct concern: that the in-sample item-level AUC of 0.995 (reference row) is partially tautological, because cobidders are by construction always-loser firms with elevated participation counts. Audit 1 holds the score fixed and swaps the label to direct-CADE defendants, returning 0.506—random discrimination, confirming the structural scope of the construct (cover bidders, not ringleaders). Audit 2 isolates the structural component by withholding cobidder firms from score construction within five cross-validation folds and re-evaluating on items containing held-out cobidders: AUC 0.891 (95% CI [0.887, 0.894]). Audit 3 evaluates the construct under temporal holdout (train 2009–2016, test 2017–2019): AUC 0.864 (95% CI [0.858, 0.870]). The screen retains discriminating power above 0.85 under both audits.

The drop of 0.10–0.13 is the pure-leakage component, and we adopt the temporal-holdout AUC of 0.864 as the conservative discriminating reference for the temporal-holdout audit in §9.3.

A placebo specification using the supply of sub-threshold always-losers as an instrument confirms that the construct's price imprint draws on the above-threshold cover-bidder population, not on generic always-loser supply. We report it in Appendix Appendix C, Table C.2.

9.3. Temporal Holdout: Generalization of the Screening Statistic

In-sample evaluation of the screening statistic confounds classification with prediction: items in the late panel contribute both to score construction and to the labels the score is evaluated against. The temporal-holdout audit disambiguates the two. Constructing the score on 2009–2016 participation only and evaluating on items in 2017–2019 yields a firm-level AUC of 0.864 (95% CI [0.858, 0.870]) against the 193 adjudicated CADE-cobidder labels. We report this as the generalization reference for the screening statistic throughout the paper.

Table E.1 reports precision, recall, and lift at five top-k cuts under both in-sample and temporal-holdout evaluation. At top-100 precision is 0.070 (lift 6.1×); at top-500 precision 0.070 (lift 6.1×); at top-1,000 precision 0.066 (lift 5.8×). The screen flags 35 adjudicated cobidder firms out of 193 at top-500 (recall 18%); at top-1,000 recall rises to 34%. The in-sample column over-states precision by approximately 50% at top-500 (0.132 in-sample versus 0.070 holdout). The inflation arises because roughly 47% of the top-500 in-sample ranking comes from 2017–2019 participation, after CADE adjudications were already underway for some of the cartels: the in-sample score is partly retrospective rather than purely prospective. We disclose the gap explicitly and report headline metrics on the holdout column.

The generalization audit reads against the screening framing of §5.3 as a diagnostic of how well the participation primitive separates cartel-adjacent from non-cartel-adjacent firms when the score is computed on data predating the labels. The drop from in-sample to temporal holdout ($\approx$ 0.10–0.13 AUC) is the pure-leakage component of the in-sample evaluation; the residual AUC > 0.85 is the structural generalization of the screening statistic.

Robustness summary. The headline price gap and the detection-regime gradient survive the threshold-multiplier sweep, two-way clustering, and the binary-versus-continuous specification audit. The four identification audits delineate the descriptive scope of β: omitted-variable bounds rule out selection-on-observables explanations of the kind the fixed effects already absorb; strict-overlap matching delivers the within-support estimand β^{ov} as the alternative empirical object whose contrast with β is the screening-value diagnostic of §7.2; the leakage decomposition isolates the structural component of the in-sample AUC; the placebo confirms the construct operates through the threshold-crossing population, not generic always-loser supply. The temporal-holdout audit reports the generalization reference. The comparison against bid-distribution detectors that the prior literature operates with is reported in the dedicated §10.

10. Screening vs. Forensic Stages: An Architectural Test

The contribution claim of §1 is structural: under incomplete observability, enforcement architecture separates an award-layer screening stage that produces candidate flags from a bid-layer forensic stage that adjudicates suspected coordination on the bid distribution. The two stages answer different questions and require different statistics. This section tests the architectural claim empirically: the two stages should discriminate the same cartel-adjacency labels at comparable levels on different data envelopes, and the combination should beat either component alone.

Figure 3 sets up the comparison. The bid layer carries five within-tender features that the Imhof–Wallimann pipeline aggregates into a discrimination score. Most enforcement environments do not preserve those features at the analytical-warehouse layer (§3); the data coarsening that drops them also drops anything else the bid distribution would carry. What survives is the award envelope: winner identity, participant identity, item code, and negotiated price. The screening statistic this paper builds operates on those four fields alone. The empirical question of this section is whether what survives is enough.

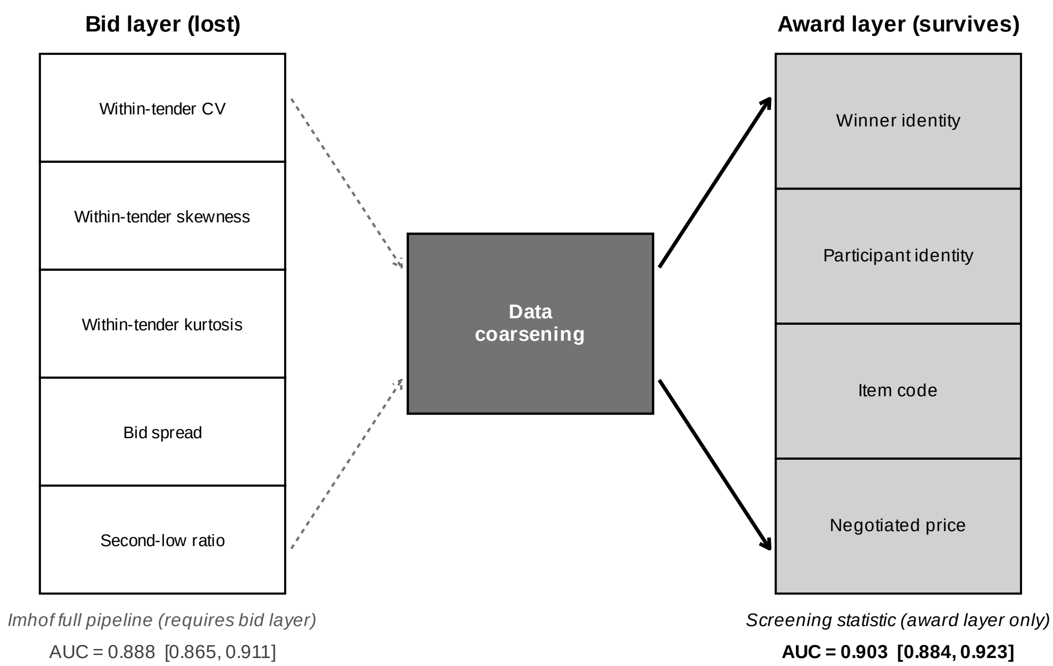

Figure 3: Bid-layer features and award-layer fields. The five bid-distribution features the Imhof–Wallimann pipeline operates on (left, dashed arrows) are dropped by the coarsening that defines most enforcement environments. The four award-record fields (right, solid arrows) survive. The boxes on the right report random-forest AUCs against 193 adjudicated CADE-cobidder labels: the two pipelines reach statistically indistinguishable discrimination on different data envelopes (Table 13).

Comparable accuracy on different data envelopes. Table 13 reports the comparison on the full always-loser pool (N = 16,779 firms, 193 adjudicated cobidder labels). The screening statistic discriminates at AUC 0.903 (95% CI [0.884, 0.923]) under the binary FL14 indicator and 0.884 ([0.860, 0.908]) under the continuous log(1 + tenders_count) score. The full Imhof–Wallimann pipeline—five within-tender bid-distribution features (coefficient of variation, skewness, kurtosis, spread, second-low ratio) (Imhof, 2019; Wallimann et al., 2023)—reaches AUC 0.888 ([0.865, 0.911]) on the same target. The two pipelines reach statistically indistinguishable discrimination on different data envelopes. The construct does not outperform the bid-distribution pipeline, and we do not claim it does; the architectural point is that comparable accuracy is achievable on a substantially thinner data envelope.

Non-redundant signal. The combined specifications in the bottom panel of Table 13 address the integration question. FL14 combined with the full Imhof pipeline reaches AUC 0.955 ([0.943, 0.967]); continuous loss intensity combined with the full Imhof pipeline reaches AUC 0.962 ([0.954, 0.969]). The combined classifiers gain 5–7 AUC points over either single construct. Restricting attention to the same-sample audit (Table E.2), where all detectors evaluate the identical subset of firms with bid microdata available, the screening statistic adds +0.035 AUC over the Imhof full pipeline alone (DeLong $p = 0.014$), and the combined model adds $+0.096 - - + 0.098$ AUC over Imhof full alone ($p < 0.001$). The award-layer screening object and the bid-layer forensic moments carry non-redundant information about the same cartel-adjacency target.

The architectural reading. The two findings together identify the sequencing structure the contribution claims. The screening statistic and the bid-distribution detectors discriminate the same labels at comparable levels on different data envelopes (the architecture treats them as substitutable when only one envelope is available); the two carry non-redundant signal when both envelopes are available (the architecture treats them as sequential when both are). Under incomplete observability the deployable architecture is therefore: an award-layer screening stage that runs on contract-record data and produces candidate environments, followed by a bid-layer forensic stage that runs the bid-distribution moment

Table 13: Imhof–Wallimann Pipeline vs. Screening Statistic: Discrimination at Different Data Costs

Model	Features required	AUC	95% CI
Participation-only (award-record level)			
FL flag (binary)	participation count > 14, win flag	0.903	[0.884, 0.923]
log(1+tenders_count)	participation count, win flag	0.884	[0.860, 0.908]
Bid-distribution (per-bid microdata)			
Imhof CV only	per-bid offer values	0.585	[0.553, 0.616]
Imhof full pipeline	5 within-tender features (CV, skew, kurt, spread, second-low)	0.888	[0.865, 0.911]
Combined			
FL flag + Imhof full	participation count + 5 bid features	0.955	[0.943, 0.967]
log(1+tenders_count) + Imhof full	participation count + 5 bid features	0.962	[0.954, 0.969]

Notes: 5-fold cross-validated AUC against 193 CADE-cobidder labels within the always-loser pool (N = 16,779). Random forests with 500 trees, otherwise default tuning. Bid-level features are aggregated to the firm level by averaging the per-tender within-tender statistic across each firm's tenders. The two top blocks compare alternative single-construct screens operating on different data envelopes; the bottom block reports the combined classifiers. The combined model gains 5–7 AUC points over either single construct, indicating non-redundant signal in the participation-only and bid-distribution information sources. The participation-only screen achieves discrimination comparable to the full Imhof–Wallimann pipeline at substantially lower data cost, which is the architectural claim of the paper: the screening stage operates on the data layer that survives most enforcement environments, and integrates with the bid-distribution forensic stage where microdata are available. We do not claim the participation-only screen *outperforms* the bid-distribution pipeline; the relevant comparison is at fixed data cost, where the two pipelines reach similar accuracy and the combination dominates.

battery on the candidates the screening stage flags, where bid microdata are recoverable for those candidates. The two stages answer different questions and require different statistics; the comparison in this section shows that they share the same cartel-adjacency target without dominating each other.

The implication for portability is concrete. Jurisdictions that preserve only the award layer can deploy the screening stage immediately, on data they already maintain. Jurisdictions that also preserve bid microdata gain the forensic stage on top, with the screening stage already constraining the pool of cases that warrant the forensic step. Reforms that mandate bid-microdata archival (§3) are best read as expanding the forensic stage rather than substituting for the screening one—a reading that reverses the standard policy intuition that bid-archival is a precondition for proactive cartel screening.

11. Limitations

The paper's empirical content has a deliberately narrow scope. This section consolidates the limitations along three dimensions: identification (§11.1), the construct itself (§11.2), and external validity (§11.3).

11.1. Identification scope

Identification is descriptive throughout the paper. Two design-based strategies that would extract a causal estimate return null at the available bandwidths: a sharp regression discontinuity at the convite/pregão statutory caps, gated by the absence of bunching in the McCrary density test, and a difference-in-differences exploiting the 2018 cap raise, gated by the failure of parallel trends. We report both failures openly.

The within-PBU coefficient 6.4% on the broad sample reverses to −9.72% under overlap-restricted matching and −30.67% under propensity-score trimming (Table 10). Neither object carries causal weight on its own; both are conditional descriptions of the same population at different scopes (§7.2). The sensitivity bounds (Cinelli and Hazlett (2020) robustness value 17.5%, Oster (2019) $\hat{\delta} = 261.6$) discipline the broad-sample association against selection on observables of the kind already absorbed by item, year, and procuring-unit fixed effects, but do not extend to the selection on unobservables that the overlap restriction surfaces. The $+3.6\%-+7.7\%$ headline range is a conditional descriptive association across estimators, not an estimate of cover bidding's causal effect on prices; the 12.6× extreme-quartile gradient is observationally consistent with the framework's detection-cost comparative static without identifying the causal channel. Under the screening framing of §5.1, this descriptive scope is sufficient: the screening object is the loser-side participation footprint where the cartel deploys it, not a counterfactual treatment effect on prices.

11.2. Construct scope

The construct flags loser-side participation patterns, not cartel members. The validation discipline established in §6—cobidders inside the always-loser stratum as the structurally appropriate object given the data layer, direct-defendant AUC of 0.49 as the design's empirical signature—defines the limit. An enforcement program that requires firm-level membership identification will need a different statistic; the screening statistic is constitutively about loser-side participation and the AUC asymmetry between cobidders (0.924) and direct defendants (0.49) is what that scope looks like in the data.

Within the always-loser stratum, the participation footprint the construct flags is observationally compatible with three non-competitive readings: cover bidding under cartel rotation, capacity-transfer signaling for subcontracting or affiliate consortia, and market-intelligence accumulation. The framework of Online Appendix A is built around the cover-bidding reading, which is the reading the strongest heterogeneity dimensions match (modal asymmetry in §8.2, detection-regime gradient in §7.3). The other two readings are not refuted; the construct flags the participation footprint these three readings share, and the empirical work is organized around the cover-bidding reading without claiming it as unique. Two finer mechanism predictions the framework generates—a cell-heterogeneity grid and a discrete first-time-FL imprint—move in the predicted direction but the data do not adjudicate them under disciplined estimation (Online Appendix C). Neither prediction is load-bearing for the screening interpretation.

The construct's behavior under deliberate adversarial adaptation is bounded but not invariant (Online Appendix G, Table F.1). The screening statistic is resilient to periodic rotation of cover bidders and to occasional wins evading the bright-line filter (AUC essentially unchanged), degrades modestly under CNPJ splitting (−3.7 points), and is substantially vulnerable to threshold-aware capping (−12.7 points); combining capping with rotation drives AUC from 0.928 to 0.632. The threshold rule is therefore the most exposed margin, and the informativeness of the screening signal in any application of the construct requires periodic recalibration alongside integration with bid-distribution methods where those data are available (§10).

11.3. External validity and portability

The empirical evidence comes from a single jurisdiction and a single time window: São Paulo's BEC, 2009–2019. The construct's portability rests on three structural features of the procurement environment, all explicit in BEC and identifiable in any candidate jurisdiction: a commodity-heavy item composition that supports within-product-code price comparisons; an electronic platform with publicly auditable participant lists; and an institutional asymmetry between the winner and loser sides of the bidding pool that makes persistent zero wins informative about strategic non-competitive participation. These are diagnostic conditions, not optional ones.

The architectural test in §10 sharpens the portability claim. Award-only enforcement environments—those that preserve participants and winners but not bid-by-bid records—are the construct's natural deployment environment; the screening stage runs on data they already maintain. Real-time-electronic environments (pregão-style auctions) expose a richer bidder-by-bidder interaction footprint that the modal contrast in §8.2 reads as the regime in which the screening signal is sharpest. Sealed-bid environments without real-time interaction are the scope-restricted boundary, not the natural deployment one.

Three settings the present paper does not test. The federal Brazilian procurement registry (ComprasNet) and other state-level registries preserve the same award-record envelope as BEC and are accessible through the same legal regime; replication is technically straightforward and substantively pending. Non-commodity settings (services, infrastructure, public works) are within the framework's logical scope but outside the empirical scope of the present work, and the within-item identifying variation that underwrites the headline price gap is not equally available in those settings. Time periods outside the sample window are likewise untested: the BEC sample begins after the platform's initial stabilization and ends before the consolidation of the new federal procurement law (Lei 14.133/2021), and the screening statistic's stability across regulatory regimes is an open empirical question. The 2021 reform consolidates pregão-style auctions as the institutional default; combined with the modal contrast in §8.2, the regulatory direction is favorable to portability rather than adversarial.

Synthesis of scope. The empirical content of the paper survives three boundary tests. Identification is descriptive but the descriptive scope is sufficient for a screening object. The construct flags loser-side participation, not cartel membership, and the validation respects the resulting label discipline. Portability is contingent on three diagnostic conditions, all observable ex ante. The paper is narrow by design; the limitations are features of the screening interpretation, not concessions against an alternative interpretation we would prefer.

12. Conclusion

The bid-rigging detection literature has been built on the assumption that bid-level data are observable. Real enforcement environments routinely violate that assumption: many electronic procurement systems preserve only the contract-award envelope, and the bid-distribution methods that work on the bid layer become inoperative on the layer that survives. The paper takes this informational mismatch as its starting point and asks what collusion-relevant statistic can be recovered when only the award layer is observable.

Endogenous loser-side participation is such a statistic. A separating-equilibrium framework with cover bidders and genuine entrants identifies $\log(1 + \text{tenders\_count})$ together with the $\text{wins} = 0$ filter as a sufficient ranking statistic for cover-bidder type given award-record data; the binary median $+ 1.5\times$IQR rule we deploy is its information-coarsening. On São Paulo's BEC (2009–2019), the construct discriminates 193 adjudicated CADE cobidders inside the always-loser stratum at firm-level AUC 0.864 under temporal holdout (train 2009–2016, test 2017–2019) and 0.748 on a strict pre-2020 contemporaneous benchmark with participation-stratified permutation null ($3.2\times$ excess over the random-matching baseline, $p < 0.001$). Against bid-distribution detectors that require per-bidder bid amounts the screening statistic reaches comparable accuracy on a substantially thinner data envelope (AUC 0.903 vs. 0.888) and adds non-redundant signal in same-sample combination ($+ + 0.035$ AUC, DeLong $p = 0.014$). Within the always-loser stratum, the cobidder validation positive class matches the modeled cover- bidder type along four of five operational predictions, with Cohen's d up to $+1.00$. AUC against the 47 direct CADE defendants in the broader BEC firm universe is 0.49; the cobidder/direct-defendant asymmetry is the design's empirical signature, since the construct recovers loser-side participation rather than winner-side identity.

The pricing evidence is descriptive corroboration. Frequent-loser presence associates with $+3.6\% - +7.7\%$ higher conditional log negotiated unit price across four estimators on the broad sample. The overlap-restricted estimate is -9.72% under ATT-reweighting and $+0.044$ under unweighted overlap restriction; the empirical decomposition in §7.2 attributes the sign reversal to ATT-weights up-weighting cells with relatively few untreated counterfactuals rather than to the overlap restriction itself (839 of 79,452 treated items, or 1.06%, are strictly dropped). We report all specifications and acknowledge the screening-value reading is one interpretation among several the data do not adjudicate. The discrimination evidence and the architectural test do not depend on overlap weighting, and they carry the contribution.

Three implications for enforcement design follow.

What survives data coarsening. The information that the bid-layer literature reads off the bid distribution is partially recoverable from a different statistic at the award layer when bid data are unavailable. The screening object the recovery delivers is a participation statistic, not a price moment, and the equilibrium-generated participation footprint that the collusive arrangement leaves on the award layer is the appropriate empirical object under coarsened observability.

Screening and forensic stages are sequenced, not parallel. The architectural test in §10 reads the screening statistic and the bid-distribution moment battery as informational complements: comparable accuracy on different data envelopes (AUC 0.903 vs 0.888 on the same cartel-adjacency target) and non-redundant signal when both envelopes are available (combined AUC 0.962, $+0.096 - - + 0.098$ over either alone). Under incomplete observability the deployable architecture is therefore an award-layer screening stage that produces candidate environments, followed by a bid-layer forensic stage that runs the bid-distribution moment battery on those candidates where microdata are recoverable. The two stages answer different questions and require different statistics.

Proactive screening does not require bid-microdata archival as a precondition. Reforms that mandate bid archival expand the forensic stage; they do not substitute for the screening one. Jurisdictions that lack the bid layer can deploy the award-layer stage on data they already maintain, and the sequencing reading reverses the standard policy intuition that bid-archival is the rate-limiting precondition for proactive cartel screening (Becker, 1968; Stigler, 1970; Baker, 2003; Harrington, 2008).

Scope and an agenda. The contribution stops at the screening stage by design. Award-record data does not contain what membership identification requires, and the cobidder population is the validation object the data layer supports. Adversarial adaptation degrades the construct's discrimination predictably (Online Appendix G), and replication rests on three diagnostic conditions—a winner/loser participation registry, an electronic platform, and an institutional asymmetry between the two sides of the bidding pool—all observable *ex ante* in any candidate jurisdiction. Three open questions follow naturally. *First,* cross-jurisdictional replication in ComprasNet, EU TED, and U.S. contracting-officer files, particularly in pregão-style real-time-electronic auctions where the modal contrast in §8.2 suggests the screening signal is sharpest. *Second,* extension beyond commodity-heavy item composition to services, infrastructure, and public works, where within-product-code price comparison is harder but the participation primitive is unchanged. *Third,* formal characterization of the optimal sequencing of screening and forensic stages under endogenous adversarial adaptation, building on the framework's deployment problem.

The conceptual content travels: wherever an enforcement environment exposes the award layer without exposing the bid layer, the question is not whether bid-distribution screens can be retrofitted onto data that does not contain them, but what participation primitive the collusive arrangement generates that survives the coarsening, and how the screening stage that reads that primitive should be sequenced before the forensic stage that requires bid-level evidence.

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